Topic 9.1 – Types of Chemical Bonds

1. Atoms form bonds to achieve a full octet. What does it mean for an atom to obtain a full octet?

2. An ionic bond is formed between two atoms, a cation and an anion. Answer the following questions regarding an ionic bond.
    a. What type of element typically forms a cation? Does the atom give or take electrons?

    b. What type of element typically forms an anion? Does the atom give or take electrons?

3. Complete the table below.

Element	# of Valence e^-	lose or take e^-?	Cation or Anion?	Ion Charge
Sodium				
Fluorine				
Aluminum				
Boron				
Nitrogen				
Sulfur				

4. A covalent bond is another method that atoms can use to achieve a full octet.
    a. How does a covalent bond differ from an ionic bond? Talk about the elements that form this bond and the behavior of the electrons.

    b. Some compounds contain single, double, and triple bonds. What is the difference between a single, double, and triple bond?

5. Complete the table below. Determine if each compound will form an ionic bond or a covalent bond.

Compound	1st Element – Metal or Non-metal?	2nd Element – Metal or Non-Metal?	Ionic or covalent bond between the atoms?
NO_2			
$MgBr_2$			
CaO			
PI_3			
SO_2			
CaI_2			

Topic 9.2 – Properties of Chemical Bonds

1. A data table has been provided to the right. Answer the questions below regarding the table.

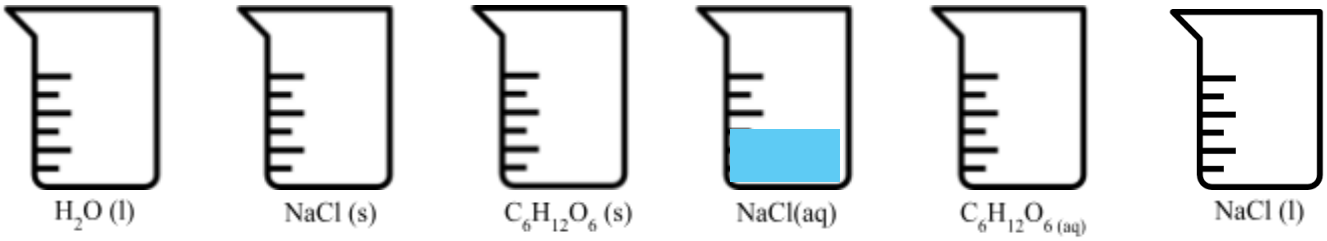
a. Identify each compound as ionic or covalent in the space provided.

b. Some compounds have a large discrepancy in the melting point. Provide an explanation using the bond found within those compounds.

Substance	Ionic or Covalent?	Melting Point	State at Room Temperature
$CaCl_2$		772°C	Solid
H_2O		0°C	Liquid
MgO		2852°C	Solid
NH_3		-77°C	Gas
SrO		2531°C	Solid
H_2S			
$C_6H_{12}O_6$		146°C	Solid

c. Based on the trend discussed in part b, do you expect H_2S to have a low or high melting point? Explain your answer by discussing the bond found within H_2S.

2. A student is testing the conductivity of certain compounds in their different states of matter. The beakers provided below show the different substances the student is testing.



a. For a substance to conduct electricity, what must be present?

b. Which substances should the student expect to conduct electrical current? Explain your answer.

c. For the solutions that you did not choose in part B, explain why you don't expect them to conduct an electrical current.

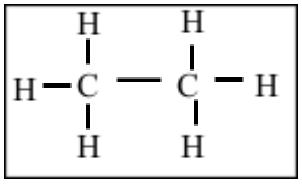
Topic 9.3 – Lewis Diagrams for Ionic and Covalent Compounds

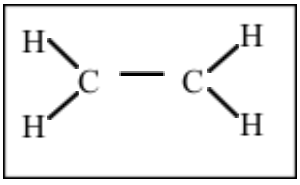
1. A student obtains bottles of potassium oxide and calcium oxide.
    a. What kind of chemical bond is found in both compounds?_____
    b. What state of matter do you expect these compounds to be in?_____
    c. The formulas for potassium oxide and calcium chloride are K_2O and CaO respectively.

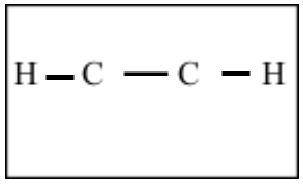
        i. Using Lewis dot diagrams, model the bonding that occurs in each compound to explain why the chemical formulas are different.

    d. The student grabs another compound that is similar to the ones mentioned above. The label has worn off, but the student knows that this compound contains lithium and nitrogen. Use a Lewis dot diagram to derive the chemical formula of this compound.

2. A student is interested in learning about the bond lengths of 3 compounds.
    a. The first diagram is complete, but the 2nd and 3rd are incomplete. Finish the diagrams by adding in the correct number of electrons and bonds.



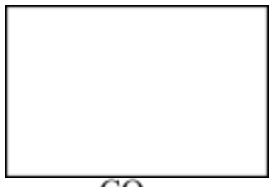




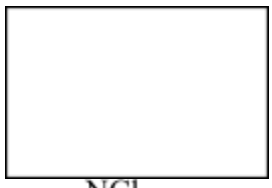
    b. Which of the carbon-carbon bonds is the shortest? Explain your answer by discussing the attraction between the protons and electrons.

3. Complete more practice below over drawing Lewis structures!

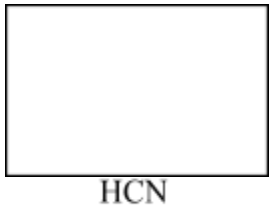

Br_2

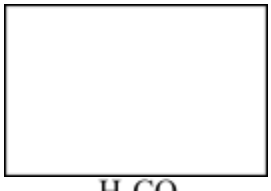

CO

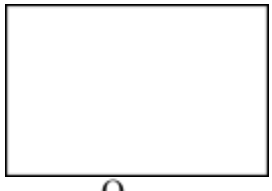

SiF_4


NCl_3


CH_3Cl


HCN


H_2CO


O_2

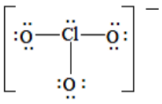
Topic 9.4 – Lewis Structures and VSEPR Theory

1. VSEPR Theory provides an explanation as to how molecules can form different shapes. In your own words, explain the big idea behind VSEPR theory and why molecules arrange themselves into specific orientations.

2. Complete the Lewis structures below and identify the shape and the angles present in each.

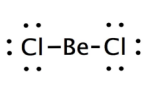
Compound	Lewis Structure/Shape/Angle(s)	Compound	Lewis Structure/Shape/Angle(s)
NO_3^{-1}	Shape: Angle:	OF_2	Shape: Angle:
PF_5	Shape: Angle:	CO_2	
SCl_6	Shape: Angle:	PH_3	Shape: Angle:

3. Identify the shape and angle of each provided Lewis Structure



Shape: _____

Angle(s): _____



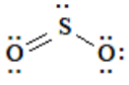
Shape: _____

Angle(s): _____



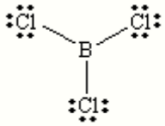
Shape: _____

Angle(s): _____



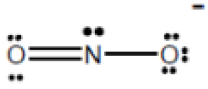
Shape: _____

Angle(s): _____



Shape: _____

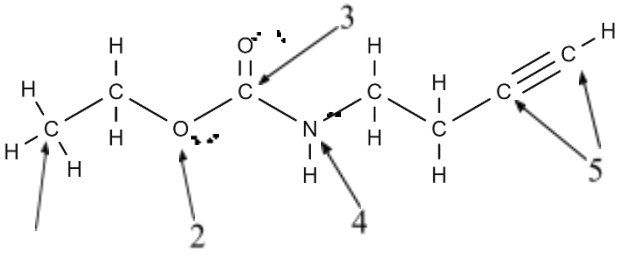
Angle(s): _____



Shape: _____

Angle(s): _____

4. The molecule provided below contains many bonds and atoms. Determine the shape around each atom indicated.



1: _____

2: _____

3: _____

4: _____

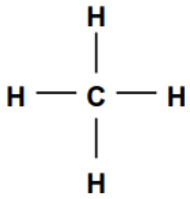
Topic 9.5 – Lewis Structures/VSEPR/Polarity

Read this first!

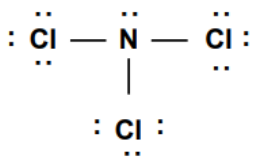
1) Carbon and hydrogen are VERY CLOSE in electronegativity values. Therefore, the carbon-hydrogen bond does NOT form a dipole. You DO NOT need to draw an arrow for a carbon-hydrogen bond.
2) The order of the top 4 electronegativities
    a. Fluorine > oxygen > chlorine > nitrogen

For the problems below, draw arrows for each bond indicated the pull of the electrons. Determine if the molecule is polar or non-polar. If the molecule is polar, assign partial charges to each element in the molecule.

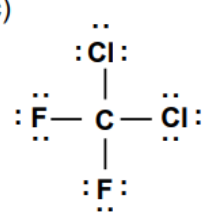
(a)



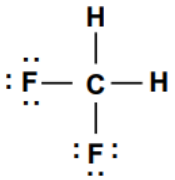
(b)



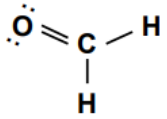
(c)



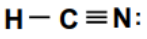
(d)



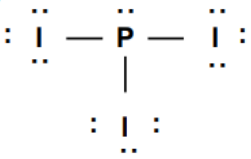
(e)



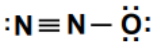
(f)



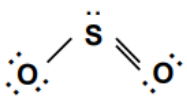
(g)



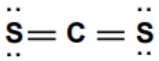
(h)



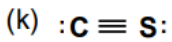
(i)



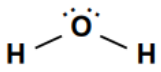
(j)



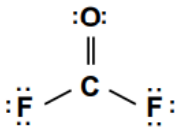
(k)



(l)



(m)



(n)



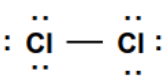
(o)



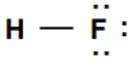
(p)



(q)



(r)



(t)

